

Observer-Relative Undifferentiated Probability Envelopes: Operational Equivalence, Ontological Underdetermination, and the First-Distinction Boundary

观察者相对的未分化概率包络：操作等价、本体未定与第一分化边界

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Abstract

This structural note freezes a temporary lower boundary in a continuing research chain on observation, admissible state spaces, and coarse-grained readout. The central proposal is deliberately modest: an ‘undifferentiated’ state is treated operationally, not as literal nothingness and not as a quantum coherent superposition. Given a candidate state set Ω and an observer/interface O specified by an allowed family of interventions and readouts, O induces an operational equivalence relation. When all candidate states lie in one equivalence class, the effective description is undifferentiated relative to O . We represent the least-committal current description by a probability envelope μ over unresolved candidates (Model A). A deeper completion in which every individual run already possesses a unique actual state λ^* (Model B) is retained as an explicit open ontological hypothesis, but is not promoted to physics unless it yields an intervention-dependent prediction distinct from Model A. We prove a simple first-distinction result: a minimal non-trivial refinement of the coarsest partition has exactly two blocks, while stressing that this binary partition is not yet a physical 0/1 ontology. We also formulate an operational non-identifiability criterion for A/B completions and locate the assumptions that any deeper model must confront in relation to contextuality, Bell constraints, and ψ -epistemic no-go results. The note does not claim a new interpretation of quantum mechanics. Its contribution is a disciplined boundary formalization: probability description is used where evidence supports only probability; deeper single-actual-state ontology is preserved as a research hypothesis; coherent quantum superposition is not assumed at the pre-differentiation layer.

Keywords: operational equivalence; undifferentiated state; probability envelope; ontological model; observer-relative state; first distinction; coarse-graining; quantum foundations

中文摘要

本文固化当前向更深层推理所达到的一个暂时边界。这里的“未分化”不被理解为绝对的“无”，也不被直接理解为量子相干叠加，而被定义为相对于某一观察接口/操作协议的不可区分性。给定候选状态空间 Ω 与观察者 O 所允许的干预和读出， O 诱导操作等价关系；若所有候选状态落在同一等价类中，则相对于 O 的有效描述是未分化的。当前采用最少承诺的概率包络 μ 作为正式操作模型（A）；同时保留“每一次过程在更深层已经存在唯一实际状态 λ^* ，只是观察者不知道”的本体补全假设（B），但只有当 B 能给出 A 所没有的独立可检验预测时，才允许升级为物理模型。本文给出最小第一次分化的二块分割结果，并明确该‘二分’不等于物理二进制 0/1。本文不宣称提出新的量子力学解释，而是固化一条研究纪律：当前证据只支持概率描述时使用 A；B 必须保留但保持未证实；量子相干叠加不被放置在当前最深层。

Frozen boundary statement

Current formal model: A = probability envelope. Open hypothesis retained: B = single actual deeper state, presently unproven. Quantum coherent superposition is not assumed at the pre-differentiation boundary.

1. Introduction

A recurring difficulty in foundational reasoning is the silent movement between three different questions: what physically exists, what an observer can distinguish, and what mathematical object is sufficient to predict future observations. These questions need not have the same answer. The present note isolates them at a deliberately low level of structure.

The immediate motivation is a research chain in which continuous relational rotation and admissible state spaces were previously separated from finite readouts. In the preceding note, an admissible state space was treated as pre-projection structure, while binary, eight-class, and twelve-sector descriptions were kept downstream as possible coarse-grained readouts [1]. Pushing one layer deeper raises a different question: before coordinates, geometry, rotation, or even a distinguished state variable are fixed, what can ‘undifferentiated’ mean in an operationally defensible way?

The answer proposed here is observer-relative. ‘Observer’ is used in the operational sense of an interface, reference system, or family of allowed interventions and readouts. No consciousness postulate is assumed. Undifferentiation is then the statement that the allowed operational resources fail to separate candidate states. This move creates a precise distinction between (i) a probability description that is empirically sufficient and (ii) a deeper claim that each run already possesses one unique actual state. We freeze both possibilities without conflating them.

2. Scope and explicit non-claims

This note is a structural research document, not a claim of new quantum physics. Its primary purpose is to freeze definitions, logical implications, and future death gates at a temporary lower boundary.

In particular, the paper does not claim that consciousness creates reality; that quantum superposition is experimentally false; that an underlying deterministic state has been established; that Bell, Kochen-Specker, or PBR-type results are evaded; that binary readout is fundamental; or that the candidate state set Ω is the final ontology of nature. Any later physical interpretation must survive independent operational tests.

3. Observer-relative operational equivalence

Let Ω denote a candidate state set. Let O denote an observer/interface characterized by a set of allowed interventions U_O , a result alphabet Y_O , and conditional statistics $P(y|s,u)$ for $s \in \Omega$, $u \in U_O$, and $y \in Y_O$.

Definition 1 (O-equivalence). Two candidate states s and s' are operationally equivalent for O , written $s \sim_O s'$, if they generate identical observable statistics under every intervention available to O :

$$s \sim_O s' \Leftrightarrow P(y|s,u) = P(y|s',u) \text{ for every } u \in U_O \text{ and } y \in Y_O. \quad (1)$$

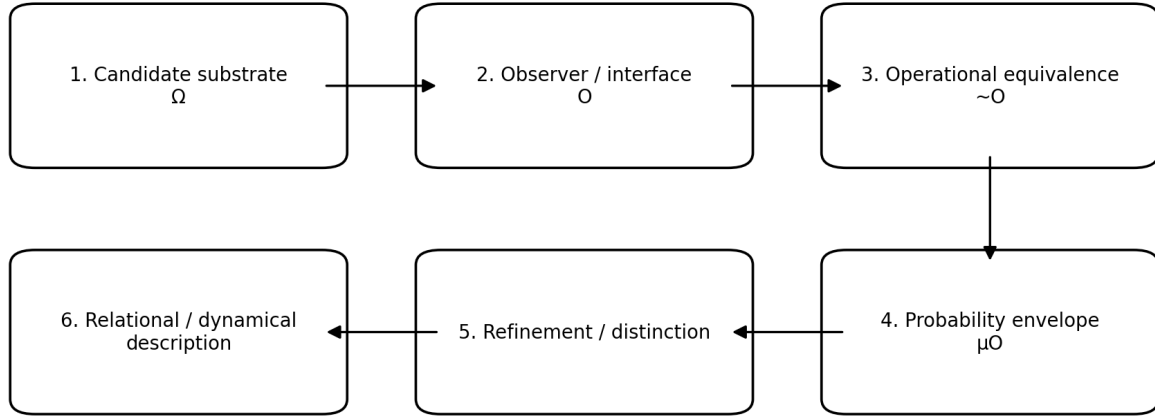
The effective state space available to O is therefore the quotient $\Omega/\sim_O$. This definition makes ‘undifferentiated’ explicitly relational rather than absolute.

Definition 2 (observer-relative undifferentiation). The candidate structure is undifferentiated relative to O when $\Omega/\sim_O$ contains a single equivalence class:

$$|\Omega/\sim_O| = 1. \quad (2)$$

A stronger observer O' may possess a larger intervention family and thereby induce a finer equivalence relation. Consequently, the same underlying candidate structure can be undifferentiated relative to O but differentiated relative to O' .

Operational hierarchy and the present lower boundary



This paper freezes the boundary at the probability-envelope / ontological-underdetermination layer.
No quantum-superposition ontology is assumed at this layer.

Figure 1. Operational hierarchy and the temporary lower boundary frozen in this note.

4. Model A: the undifferentiated probability envelope

At the present boundary, the least-committal state description is a probability measure μ_O over unresolved candidates. For a finite $\Omega = \{s_i\}$, write $p_i = \mu_O(s_i)$, $p_i \geq 0$, and $\sum_i p_i = 1$. For a continuous candidate space, μ_O is a probability measure on a specified measurable structure.

We call (Ω, μ_O) an undifferentiated probability envelope when the candidate distinctions represented inside Ω are not operationally resolved by the current observer/interface. The word ‘envelope’ indicates that several unresolved possibilities are represented together; it does not imply simultaneous physical coexistence of all possibilities.

This point is crucial. A probability envelope is not yet a statement about whether one candidate has already become actual, whether the event is objectively stochastic, or whether the probabilities are only epistemic. Model A intentionally withholds that deeper commitment. Its advantage is not merely rhetorical simplicity: it is the minimal state description required when empirical evidence determines probabilities but does not identify an underlying completion.

5. The first-distinction boundary

The observer-relative formulation permits a small but exact structural result. Consider the lattice of partitions of Ω . The coarsest partition is $P_0 = \{\Omega\}$. A distinction corresponds to refining P_0 into more than one block.

Proposition 1 (minimal non-trivial refinement). Any minimal non-trivial refinement of P_0 has exactly two blocks.

Proof. Suppose a minimal non-trivial refinement had at least three blocks $A, B, C, \dots$. Merging any two blocks, for example $B \cup C$, produces a partition strictly finer than P_0 but strictly coarser than the assumed refinement. This contradicts minimality. Hence a minimal non-trivial refinement has exactly two blocks. $\square$

The result explains why a binary code can appear at the first minimal act of distinction. It does not establish that nature is ontologically binary. ‘Two equivalence classes’, ‘two spatial dimensions’, and ‘a 0/1 readout alphabet’ are different notions and must remain separated.

6. Model B: a single actual state as an open ontological completion

A deeper hypothesis may be added to Model A: each individual run possesses one actual candidate $\lambda^* = \lambda$ before the observer resolves it. The probability envelope μ then represents incomplete knowledge, uncertainty over preparation conditions, or an effective ensemble of such actual states.

We retain this as Model B because it captures a meaningful foundational possibility that Model A deliberately leaves undecided. However, B is not presently established by the probability data alone.

The relationship can be written schematically as:

$$\text{deeper actual state } \lambda^* \rightarrow \text{observer-limited information} \rightarrow \text{probability envelope } \mu_O. \quad (3)$$

Different ontological completions can therefore map to the same operational probability model. This is an underdetermination problem, not a reason to delete B.

7. Operational non-identifiability of A and B

Let Model A and Model B generate conditional predictions $P_A(y|u)$ and $P_B(y|u)$ for every allowed intervention $u \in U_O$.

Proposition 2 (operational non-identifiability). If

$$P_A(y|u) = P_B(y|u) \text{ for every } u \in U_O \text{ and every } y, \quad (4)$$

then no experiment restricted to U_O can distinguish A from B.

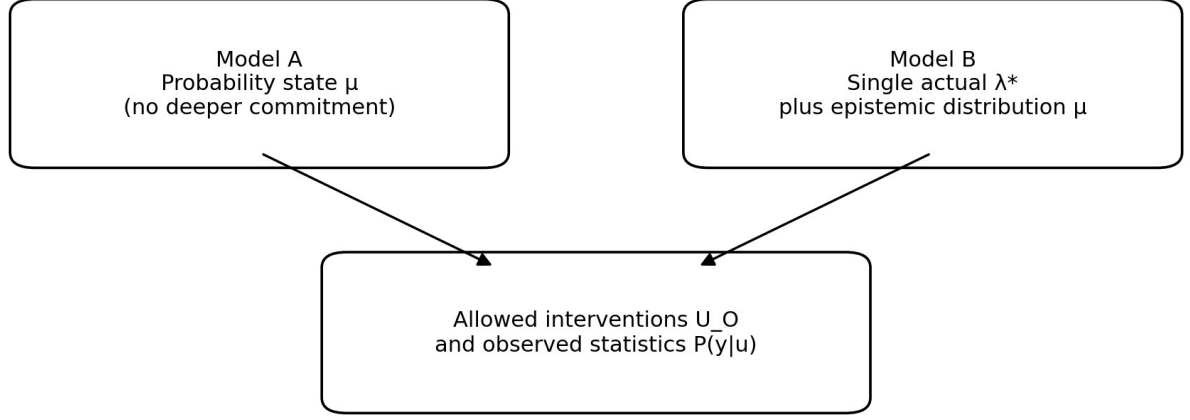
Proof. Every experiment available to O is, by definition, a selection of $u \in U_O$ followed by observation of y . Equation (4) makes the full outcome distribution identical for the two models for every allowed selection. Therefore every statistic computed from such experiments has the same sampling law under A and B. $\square$

Equation (4) is simple, but it supplies an important research discipline. A deeper ontology becomes empirical only after it produces an operational split. A sufficient empirical upgrade criterion is the existence of an intervention u^* and result y^* such that

$$P_A(y^*|u^*) \neq P_B(y^*|u^*). \quad (5)$$

Without such a split, B can be retained as an interpretive or ontological completion, but it cannot be reported as an experimentally established layer.

A/B ontological underdetermination and the empirical split criterion



If $PA(y|u)=PB(y|u)$ for every allowed u and y , A and B are operationally indistinguishable.

B becomes empirical only if a new intervention u^* produces a frozen prediction difference.

Figure 2. Model A and Model B are empirically distinct only after an intervention produces different observable statistics.

8. Why quantum coherent superposition is not placed at this boundary

The present work deliberately does not identify the undifferentiated probability envelope with a coherent quantum superposition. These are mathematically different structures.

A probability envelope lives in a probability simplex: $p_i \geq 0$ and $\sum_i p_i = 1$. A quantum pure state, by contrast, involves complex amplitudes α_i with relative phases, $|\psi\rangle = \sum_i \alpha_i |i\rangle$, together with an inner-product structure and phase-sensitive effects. The interference terms have no generic counterpart in an ordinary probability distribution.

This does not deny experimentally observed quantum coherence. It only fixes its location in the present research chain: quantum coherence is a later effective/operational structure that any deeper completion must reproduce, rather than an assumption used to define the pre-differentiation layer itself.

Accordingly, ‘the system is in one actual deeper state’ and ‘the observer has a probability envelope’ remain a hypothesis pair to be tested against phase-sensitive interference, contextuality, and entanglement statistics.

9. Constraints from quantum foundations

Any attempt to promote Model B from an open completion to a physical theory must confront established no-go results. Bell’s theorem rules out broad classes of local hidden-variable completions that satisfy the relevant factorization assumptions [2]. Kochen-Specker contextuality and its operational generalizations constrain noncontextual value assignments and noncontextual ontological models [3,4]. Pusey, Barrett, and Rudolph showed that, under additional assumptions including a form of preparation independence, simple ψ -epistemic models with overlapping ontic distributions face strong restrictions [5].

These results do not by themselves force a single universal metaphysical interpretation of the quantum state, but they sharply narrow the space of viable B-type models. A surviving deeper completion may need contextual, nonlocal, relational, retrocausal, stochastic, or otherwise nonclassical structure; which assumptions are relaxed must be stated explicitly rather than hidden in language.

Operational and general-probabilistic frameworks provide a useful neutral comparison language because they separate preparations, transformations, measurements, and observed probabilities from later ontological choices [4,6]. This is the role adopted here.

10. Relation to the pre-projection admissible-state paper

The previous paper [1] froze a pre-projection layer in which admissible states, flows, trajectories, and observation maps were separated from finite readout alphabets. The present note does not replace that result. It pushes one conceptual step earlier by asking how an observer-relative probability description and a first distinction can be defined before committing to a specific continuous coordinate representation.

The updated working chain is:

candidate/deeper substrate → observer/interface → operational equivalence → probability envelope → refinement/distinction → relational/admissible structure → dynamics/flow → projection/coarse-graining → finite readout. (6)

Equation (6) is a research map, not a derivation of nature. Each arrow requires its own mathematical and empirical gate. Figure 1 records the temporary boundary frozen by the present paper.

11. The 0/12 closure analogy and its boundary

A recurring intuition in the wider research program is that labels 0 and 12 may denote the beginning and endpoint of a closed cycle. The present framework clarifies two distinct possibilities. If 0 and 12 are related by periodic boundary identification, 0~12, they are two labels of one effective point. If deeper states s_0 and s_{12} are distinct but a projection maps them to the same readout, $\Pi(s_0)=\Pi(s_{12})$, they exhibit observational aliasing. Neither case, by itself, is a quantum coherent superposition.

The useful common motif is many-to-one equivalence: multiple labels, histories, or candidate states can correspond to one effective class under a chosen observation structure. This motif links the 0/12 closure analogy to the present undifferentiated-envelope formalism without making an unsupported quantum claim.

12. Research death gates beyond the present boundary

The current reasoning has reached a useful temporary boundary. Further progress should not proceed by adding unconstrained ontology. It should proceed through explicit gates.

Gate B1 - Single-system interference: can a model with one actual deeper state per run reproduce phase-sensitive interference without simply storing the complete quantum state as hidden data?

Gate B2 - Contextuality: if the deeper response depends on the measurement context, specify that dependence and compare it with operational contextuality tests.

Gate B3 - Bell: for composite systems, state exactly which Bell assumptions are retained or relaxed and derive the corresponding correlation bounds.

Gate B4 - Identifiability: find an intervention for which Model B predicts a distribution that cannot be reproduced by Model A within its declared state space, or establish that no such intervention exists in the chosen task class.

Gate B5 - Complexity: even if a deeper completion reproduces quantum statistics, test whether it is genuinely simpler, more explanatory, or more predictive than the effective quantum representation it is intended to underlie.

Gate D1 - First distinction to relational geometry: identify the minimal extra structure that turns a two-block distinction into ordered, metric, topological, or continuous degrees of freedom. This is distinct from the B gates and can be pursued independently.

13. Discussion: what has and has not been reached

The main conceptual gain is a separation of layers. ‘Undifferentiated’ is not treated as absolute nothingness. It is a statement about unresolved distinctions relative to an operational interface. The associated probability envelope is

sufficient at the current evidence level. The stronger statement that one candidate is already actual in every run is retained, explicitly labelled, and postponed to future discriminating tests.

This dual-track treatment avoids two opposite errors. The first is to erase a potentially meaningful deeper hypothesis merely because current experiments do not identify it. The second is to present an empirically underdetermined ontology as if it had already been established. Model A and Model B are therefore not competing slogans; they occupy different epistemic statuses.

The paper also marks a methodological limit. No further ‘lower’ layer is claimed merely by verbal descent. If future reasoning pushes beneath the present boundary, it should introduce either a mathematically necessary structure or a new operational distinction. Otherwise the lower-layer language remains philosophical scaffolding rather than physics.

14. Conclusion

We have formalized a temporary lower boundary in a structural research chain. An observer/interface induces operational equivalence on a candidate state set. Complete equivalence yields an observer-relative undifferentiated effective state. The least-committal current representation is a probability envelope μ (Model A). A single-actual-state completion λ^* (Model B) is retained as an open, unproven hypothesis. A and B are empirically distinguishable only if some allowed or newly introduced intervention produces different observable statistics. A minimal first distinction is binary at the level of partition refinement, but this does not establish a binary physical ontology. Quantum coherent superposition is not assumed at this boundary, although any deeper model must ultimately reproduce quantum interference and other established phenomena.

The immediate research program is therefore twofold: preserve the A/B split without premature closure, and search for the smallest operational or mathematical structure capable of pushing the boundary further. Until such a structure is found, the present note is intended to serve as a frozen checkpoint rather than as a final ontology.

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Status statement

This document is a structural preprint / formal research note. It freezes a temporary research boundary and does not claim experimental validation of a deeper ontology. Model B is intentionally retained as an open hypothesis rather than silently removed or promoted to fact.